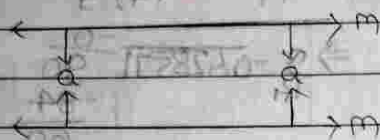


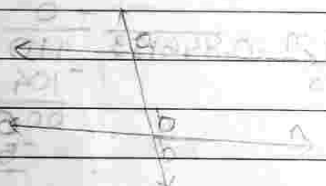
2. Parallel lines and transversal



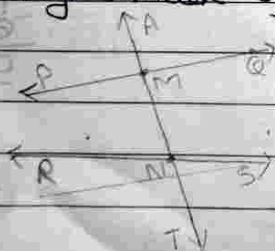
Parallel lines :- The lines which do not intersect each other are called parallel lines.

The line m and line n are parallel line. It is written as line $m \parallel$ line n .

Transversal :- If a line which intersects given two lines in two distinct points. Then the line is called transversal. The line p is the ~~intersect~~ transversal of line m and n .



Angles made by a transversal :-



If a transversal intersects two given lines. Then we get the point of angles.

Pairs of Angles :-

① Corresponding Angles :-

- $\angle AMP$ and $\angle MNR$
- $\angle PMN$ and $\angle RNT$
- $\angle AMQ$ and $\angle MNS$
- $\angle QMN$ and $\angle SNT$

② Interior angles :-

- $\angle PMN$ and $\angle MNR$
- $\angle QMN$ and $\angle MNS$

③ Alternate Angle :-

(i) Interior alternate angle :-

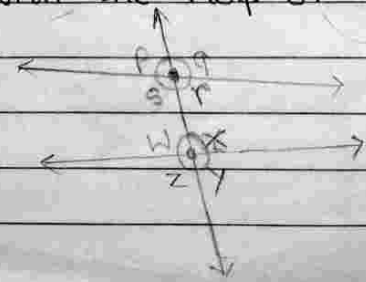
- $\angle PMN$ and $\angle MNS$
- $\angle QMN$ and $\angle RNM$

(ii) Exterior alternate angle :-

- $\angle AMP$ and $\angle TNS$
- $\angle AMQ$ and $\angle RNT$

Practice Set 2.1

1. In the adjoining figure, each angle is shown by a letter. Fill in the boxes with the help of figure.



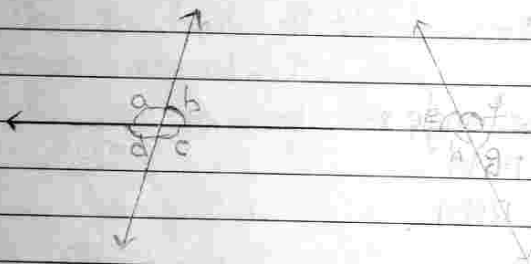
Corresponding angles.

- (1) $\angle p$ and $\angle w$ (2) $\angle q$ and $\angle x$
 (3) $\angle r$ and $\angle y$ (4) $\angle s$ and $\angle z$

Interior alternate angles.

- (5) $\angle s$ and $\angle x$ (6) $\angle w$ and $\angle y$

2. Observe the angles shown in the figure and write the following pair of angles.



(1) Interior alternate angles.

- (1) $\angle c$ and $\angle e$ (2) $\angle b$ and $\angle h$

(2) Corresponding angles.

- (1) $\angle a$ and $\angle e$ (2) $\angle b$ and $\angle f$

(3) Interior angles

- (1) $\angle c$ and $\angle h$ (2) $\angle b$ and $\angle e$

Now I know.

When two parallel lines are intersected by a transversal, the angles formed in each pair of

- ① Corresponding angles are congruent.
- ② Alternate angles are congruent.
- ③ Interior angles are supplementary.

Practice Set 2.2

1. Choose the correct alternative.

- (1) In the adjoining figure, if line $m \parallel$ line n and line p is a transversal then find x .

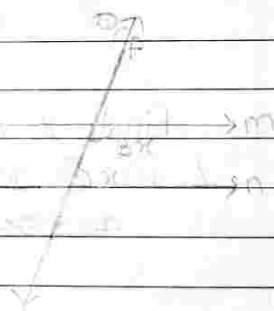
(A) 135° (B) 90° (C) 45° (D) 40°

$$3x + x = 180$$

$$4x = 180$$

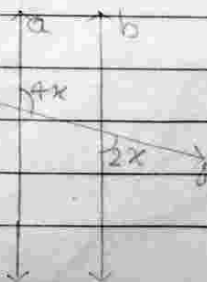
$$x = \frac{180}{4}$$

$$x = 45$$



- (2) In the adjoining figure, if line $a \parallel$ line b and line l is transversal then find x .

(A) 90° (B) 60° (C) 45° (D) 30°



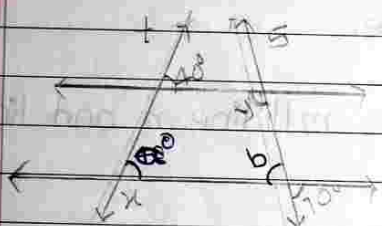
$$\Rightarrow 4x + 2x = 180$$

$$6x = 180$$

$$x = \frac{180}{6}$$

$$x = 30$$

2. In the adjoining figure line $p \parallel$ line q , line t and line s are transversals. Find measure of $\angle x$ and $\angle y$ using the measures of angles given in the figure.



Find x Value.

$$\Rightarrow \angle 40^\circ \text{ \& } x = \text{corresponding angle.}$$

$$\angle 40^\circ + x = 180$$

$$\therefore x = 180 - 40$$

$$\therefore 40 + x = 180$$

$$x = 180 - 40$$

$$= 140$$

$\Rightarrow$ Find y value

Opposite angle = 70° and b

$$\therefore b = 70$$

$$b + y = 180$$

$$70 + y = 180$$

$$y = 180 - 70$$

$$= 110^\circ$$

3. In the adjoining figure, line q || line m . Find measure of $\angle a$, $\angle b$ and $\angle c$, using the measures of given angles. Justify your answer.

Opposite $\angle$ of a is $\angle c$

$$\therefore 80 + a = 180 \quad [\text{Interior Angle}]$$

$$a = 180 - 80$$

$$\angle a = 100^\circ$$

Corresponding $\angle$ of a is 100°

$$\therefore 100 + b = 180 \quad [\text{Linear pair}]$$

$$b = 180 - 100$$

$$\angle b = 80$$

Corresponding $\angle$ of a is 100°

$$\therefore 100 + c = 180 \quad [\text{Linear pair}]$$

$$c = 180 - 100$$

$$\angle c = 80$$

4. In the adjoining figure, line a || line b . Line l is a transversal. Find the measures of $\angle x$, $\angle y$, $\angle z$ using the given information.

$\Rightarrow$ Corresponding angle 105° & x

$$\therefore \boxed{x = 105^\circ}$$

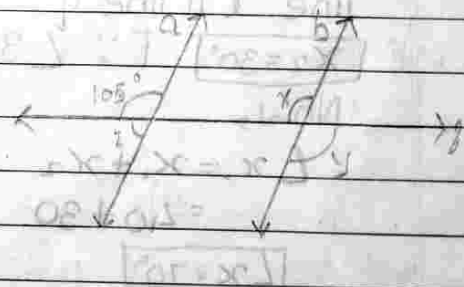
Opposite angle x & y

$$\therefore x = 105^\circ = \boxed{y = 105^\circ} \quad [\text{Linear pair}]$$

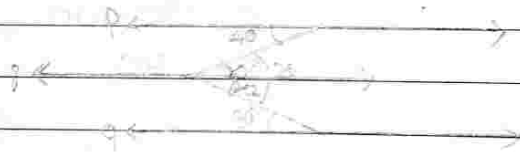
$$105 + z = 180 \quad [\text{Interior Angle}]$$

$$z = 180 - 105$$

$$\boxed{z = 75}$$



5. In the adjoining figure, line $p \parallel l \parallel$ line q .
Find $\angle x$ with the help of the measures given in the figure.



$\Rightarrow$ line $p \parallel$ line l .

$\boxed{x_1 = 40^\circ}$ [$\because \angle 40^\circ$ & x , Alternate Angle]

line $l \parallel$ line q

$\boxed{x_2 = 30^\circ}$ [$\because \angle 30^\circ$ & x , Alternate Angle]

Now,

$$\angle x = x_1 + x_2$$

$$= 40 + 30$$

$\boxed{\angle x = 70^\circ}$